Sahaj Bhadja

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Summary

Application-focused Software Engineer with 2+ years of experience in **technical requirements**, **testing and troubleshooting**, performance analysis, cross-functional delivery, and **clear engineering documentation** for complex systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **validation, reconciliation, and error handling**, cloud storage, CI/CD releases, and production monitoring supporting dependable technical operations and stakeholder workflows.
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and role-based isolation for a multi-tenant analytics platform supporting 500+ users reliably.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, secure ingestion, AWS RDS, and CI/CD workflows.
- Maintained 90%+ scoring consistency across 10K+ submissions through **structured test specifications**, fallback paths, persistent results, and reviewer controls supporting reproducible evaluation workflows and clear technical issue resolution.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **quantitative evaluation pipeline** over 100K+ autonomous-driving logs with controlled experiments and reproducible performance analysis across diverse scenarios.
- Improved closed-loop planning stability by 15% through a **simulation-based validation model** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex scenarios reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented services**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments under sustained traffic.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating test reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Pandit Deendayal Energy University | *Research Fellowship* | *India* **January 2023 – May 2024**
- Designed a **distributed authentication architecture** spanning users, IoT devices, and role-based access scenarios, evaluating immutable identity records and credential models while documenting requirements, assumptions, and technical trade-offs.
- Published peer-reviewed security research after analyzing protections against replay, phishing, man-in-the-middle, and credential-theft attacks, translating complex findings into technically defensible documentation and presentations for multidisciplinary IEEE audiences.

Skills

Languages: Python, C++, Java, SQL, JavaScript, TypeScript, Bash
Testing & Analysis: Test Planning, Test Specifications, Unit Testing, Integration Testing, Automated Testing, Regression Testing, Debugging, Root-Cause Analysis
Data & Visualization: PostgreSQL, MySQL, AWS RDS, Data Modeling, Query Optimization, Matplotlib, Data Visualization
Systems & Tools: Linux, Windows, Git, GitHub, Jenkins, Docker, CI/CD, REST APIs, Object-Oriented Programming
Documentation & Delivery: Requirements Analysis, Technical Documentation, Test Reports, Technical Specifications, Technical Presentations, Performance Metrics, Issue Tracking, Software Development Lifecycle
Collaboration: Cross-Functional Collaboration, Stakeholder Communication, Problem Solving, Technical Support, Project Coordination

Projects

PIC Evaluation and Reporting Suite - Design Concept
- Designed a **Python evaluation-suite concept** to ingest laboratory measurements, compare performance against configurable specifications, flag anomalies, generate traceable test reports, and produce customer-ready charts, summaries, and technical documentation.

Customer Requirements and Issue Tracker - Design Concept
- Designed a **technical-engagement workflow concept** linking customer requirements, product capabilities, test evidence, owners, decisions, and timelines, with searchable specifications, review checkpoints, issue histories, and exportable application-note drafts.

CPU-Scheduling Simulator | Live
- Built a **CPU-scheduling simulator** implementing FCFS, SJF, SRTF, Priority, and Round Robin algorithms with configurable workloads, timeline visualization, comparative metrics, and deterministic validation across scheduling scenarios reliably.